Sooryabalan Murugesan

MSc Data Science

Contact

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Skills

Programming: Python, Java, MATLAB
Scientific Computing: NumPy, SciPy, Matplotlib, Astropy, emcee, CosmoloPy
Other: Mathematica, LaTeX, Arduino

Outreach & Leadership

TIFR Outreach Activities
Science Communicator
2022 - 2025

Anantham Club Activities
Maths Communicator
2019 - 2022

Profile

Data Science student at Chennai Mathematical Institute (CMI) with an MSc in Physics from Tata Institute of Fundamental Research (TIFR). Experienced in statistical modelling, Bayesian inference, numerical methods, and scientific data analysis through research in cosmology and gravitational-wave physics. Strong mathematical and quantitative background with hands-on experience in Python and scientific computing.

Education

Chennai Mathematical Institute — **Master of Science**
Data Science, 2026 - Present — CGPA: -/10
Key Courses - Probability and Statistics with R, Mathematical Methods - Analysis, Programming and Data Structures with Python

Tata Institute of Fundamental Research — **Master of Science**
Physics, 2022 - 2025 — CGPA: 7.6/10
Key Courses - Probability & Statistics, General Relativity, Cosmology II, Astronomy & Astrophysics I & II, Particle Physics

Amrita Vishwa Vidyapeetham — **Bachelor of Science**
Physics, 2019 - 2022 — CGPA: 8.74/10

Projects

Handling unknown systematic errors in Quasar Luminosity Function measurements

MSc Thesis / Cosmology | August 2024 - July 2025

Guide: Prof. Girish Kulkarni (TIFR)

Based on Kulkarni et al. (2019), which highlights the presence of unknown systematics in QLF estimates. This project developed a statistical framework to model and marginalize over such systematics, providing an updated and comprehensive QLF estimate using the largest quasar dataset to date.
Tools: Python

Analytic Fourier Domain Inspiral Templates for Spinning Binaries in Eccentric Orbits

Theoretical Astrophysics| August 2024 - Jan 2025

Guide: Prof. Achamveedu Gopakumar (TIFR)

The project has derived Fourier domain inspiral templates for eccentric binaries with 1.5PN spin corrections for enhanced gravitational wave data analysis. This project works heavily on perturbation theory (PN approximation), and other approximations such as, Bessel Expansion, and Small Eccentricity limit.
Tools: Mathematica

Positions of Responsibility

Teaching Assistant - Numerical Methods, TIFR
Graded assignments and examinations; conducted tutorials on efficient algorithms and computational methods.

Cultural Secretary, TIFR Students' Society (TSS)
Elected Member of TSS, Organised 10+ cultural events throughout the year

Achievements

National Probability Challenge | Techfest, IIT Bombay x Jane Street - 2026
Qualified for 2nd Round

Common Admission Test (CAT) - 2025
Percentile - 97.52

Test Of English as a Foreign Language (TOEFL) - 2024
Total Score - 100

Joint Admission test for Masters (JAM) - 2022
All India Rank - 114

Characterising Hyperbolic Passages of Spinning Black Holes
Theoretical Astrophysics| Jan 2024 - July 2024
Guide: Prof. Achamveedu Gopakumar (TIFR)
The project has derived a 1.5PN accurate Keplerian parametric solution for spinning compact binaries in hyperbolic orbits and computed the radiated gravitational wave energy. Additionally, it developed an 1.5PN accurate GW energy radiated model, which can enhance the analysis of spinning black hole binaries with detectors like LIGO and LISA.
Tools: Mathematica

Study of Stellar Structure Equations with Quark-Gluon Matter Inside the Core of Neutron Star
BSc Thesis / Theoretical Astrophysics| Jan 2022 - June 2022
Guide: Dr. Sreekanth V. (Amrita Vishwa Vidyapeetham)
This project derived the TOV equations and numerically solved them using the RK4 method with both the MIT Bag Model and the Polytropic Model as equations of state, reproducing the results of Reddy et al. (2004).
Tools: MATLAB

Scholarships

Junior Research Fellowship (JRF)
at TIFR | Aug 2022 - July 2025
- Awarded through the competitive Integrated MSc–PhD programme at TIFR to support postgraduate research in physics.
- Funded by the Department of Atomic Energy, Government of India, with research stipend and annual contingency support.

Conferences & Workshops

Young Astronomers' Meet (YAM)
at TIFR | 18 - 21 March 2025

Pune-Mumbai Cosmology and Astro- Particle (PMCAP) Meeting – Chapter 4
at IIT Bombay | 7 - 8 February 2025

Pune-Mumbai Cosmology and Astro- Particle (PMCAP) Meeting – Chapter 3
at TIFR | 13 - 14 September 2024

GW-EM-Nu-2023 conference
at TIFR | 20 - 24 November 2023